The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “internal energy” and “work” to explain your answer. Sentence starters are optional.

internal energy: energy stored in a system’s microscopic motion/interactions

work: energy transferred by a force along a displacement

surroundings: everything outside the chosen system

01 Warm-up: bring back an earlier idea

A **0.3 kg** vapor releases **100000 J** while condensing at constant temperature. **Find latent heat** and signed Q for the sample.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives **90 J** of heat and does **30 J** of work on its surroundings. With ΔU=Q-W, internal energy increases by **60 J**.

Your turn: Draw labeled arrows crossing a system boundary and **mark the 60 J** stored increase.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

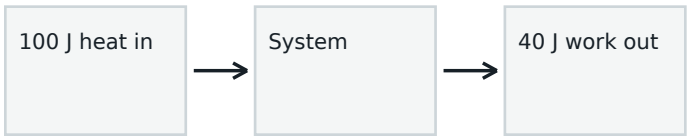
Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A system absorbs **100 J** of heat and does **40 J** of work. **Draw the energy arrows** and **find its internal-energy change**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A gas loses **50 J** of heat while a piston does **60 J** of work on it.
1. **Label both transfers** on the system boundary.
2. Use $\Delta U = Q - W_{by}$, with work done by the gas positive, to calculate ΔU.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

The first law

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why does the phrase “work is **60 J**” need a direction or sign convention?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart at **12 m/s** hits an identical cart at rest. Final velocities are **5** and **7 m/s**. **Check momentum** and energy; classify the collision.
Show the energy changes and transfers; **label units**.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial | **Transfers** | **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A process has ΔU=**-15 J** and W_{by}=**25 J**.
1. **Find Q** and say whether heat enters or leaves.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A system loses **20 J** of heat and has **50 J** of work done on it. Using work-by-system as positive, find Q, W and ΔU. Then reverse both transfers. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3